

Real-Time Traffic Anomaly Detection with MLOps

Abstract

An end-to-end MLOps pipeline that detects traffic anomalies from Estonian census data by processing hourly measurements of volume, speed, and vehicle types, enabling proactive traffic management through automated model deployment and monitoring.

Workflow

The system continuously ingests new data files, extracts temporal and traffic features, applies an anomaly detection model, exposes predictions through an API, and monitors performance to trigger retraining with fresh data and operator feedback.

Purpose	Technology
Baseline training	Scikit-learn Isolation Forest
Experiment tracking	MLflow
Scheduling	Prefect (hourly job)
Anomaly scoring	FastAPI microservice
Drift detection	Evidently AI
Dashboard	Streamlit + Folium (map) (maybe)
Storage	Parquet
Deployment	Docker
Version control	Git, DVC, MLflow

Data: <https://andmed.eesti.ee/datasets/liiklusloenduse-andmed>

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